

WASP-104b

R-2652

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_150221 · created 2026-08-29T23:31:44+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457074.9201 +/- 0.00061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.034
Transit depth (Rp/Rs)^2	2.03 +/- 0.98 [%]
Orbital Inclination (inc)	84.1 +/- 2.08
Airmass coefficient 1 (a1)	1.065 +/- 0.017
Airmass coefficient 2 (a2)	-0.043 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.58 %
Best Comparison Star	#3 - [171, 284]
Optimal Aperture	3.02
Optimal Annulus	8.59
Transit Duration (day)	0.073 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.142 $\pm$ 0.034 vs 0.1214 (deviation 0.61 σ)
Duration fit vs published	0.073 d vs 0.0742 d (deviation 0.06 σ)
Mid-time O-C	-1.9 min
Depth SNR	4.2
Residual scatter	1.58 %

Light curve

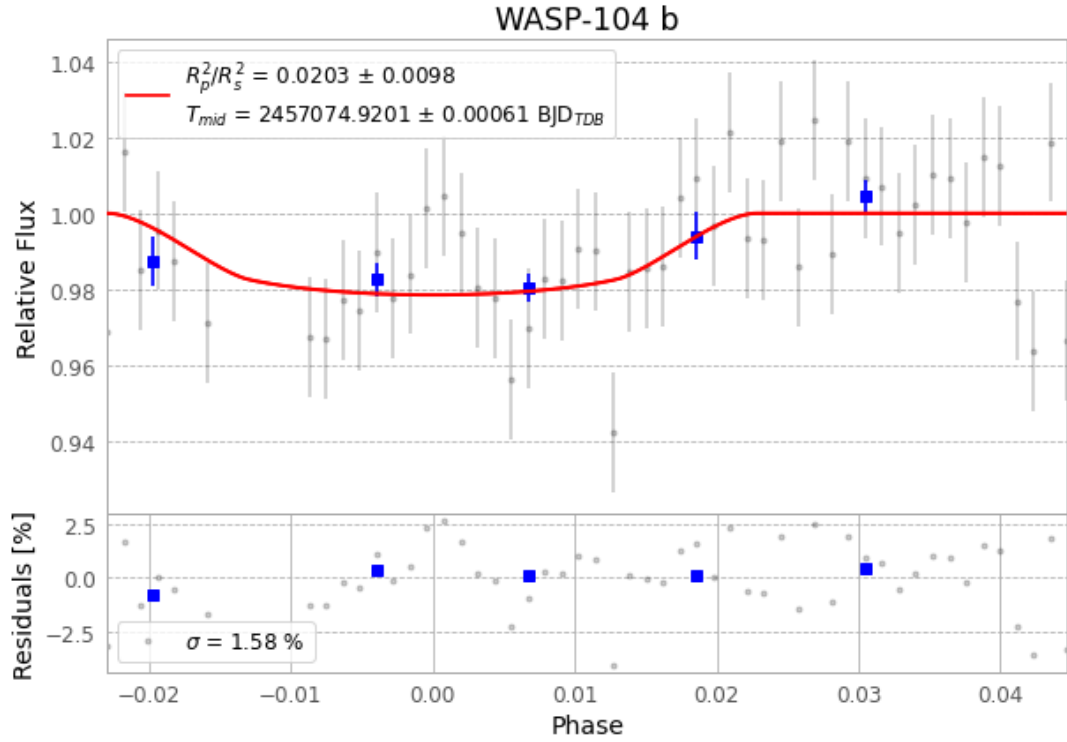


Figure 1 — FinalLightCurve_WASP-104 b_2015-02-21.png (SHA-256 98fdf2b9793364b1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [157, 241], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3b737896aeb47557...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

58 FITS frames (38 MB), WASP-104150221085712.FITS → WASP-104150221114812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-150221.log (0a8d83e00461...), FinalLightCurve_WASP-104 b_2015-02-21.png (98fdf2b97933...), FinalLightCurve_WASP-104 b_2015-02-21.csv (76ca95121616...)

Compute resources

Wall time 128.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 23:31Z.